

Atlas P-100 Overview

The Atlas P-100 is Northwind's flagship autonomous picking robot, launched in 2021. It carries payloads of up to 15 kg and travels at a top speed of 2.5 m/s in mixed pedestrian environments.

Over 3,000 units are deployed across Europe and North America.

Battery and Charging

The P-100 uses a lithium iron phosphate (LiFePO₄) battery pack rated for 8 hours of continuous operation. A fast-charge dock restores 80% capacity in 45 minutes.

Battery packs are hot-swappable; a trained operator can exchange a pack in under 90 seconds.

Sensing and Navigation

Navigation combines six stereo camera pairs with two solid-state lidar units with a range of 100 meters. The chassis is rated IP54 for dust and splash resistance.

Localization accuracy is within 2 centimeters using pre-mapped warehouse floor plans.

Software Platform

The P-100 runs on a ROS 2 based software stack. Fleet coordination is handled by Northwind's Hive fleet manager, which supports up to 500 robots per site.

Over-the-air software updates are released on a monthly cadence.

Safety Systems

The P-100 complies with the ISO 3691-4 standard for driverless industrial trucks. The emergency stop system halts the robot within 250 milliseconds.

A protective safety zone of 1.2 meters is enforced around detected humans; entering it triggers an immediate slowdown.